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1. Background: Topological Data Analysis

Traditional financial models predominantly rely on simple pairwise metrics that fail to capture the complex, high-dimensional topologies of financial markets, a fundamental limitation addressed by Topological Data Analysis (TDA) [1] which employs simplicial complexes to extract the underlying geometric skeleton of systemic risk, drawing direct inspiration from recent applications of persistent homology in digital and global equity markets [2, 3]. **Specifically, we emulate the topological data analysis approach presented in [2] on the market data for the COVID-19 "Black Monday" (March 16th, 2020) across a state space of 20 key stocks from the S&P 500 and cryptocurrency exchanges.** The primary objective is to identify and quantify the topological phase transitions that immediately precede these catastrophic market crashes, ensuring analytical robustness and manifold stability to construct a comprehensive representation of global market dynamics.

2. Data Sourcing, Preprocessing & Metric Embedding

1. Data Sourcing & Preprocessing

We source daily pricing data for the S&P 500 constituents and cryptocurrency markets via the **yfinance** API. Our data pipeline architecture is inspired by methodologies discussed in [4].

Analysis Timeframe: January 1st, 2019 – December 31st, 2021 (encompassing the COVID-19 crash).

To address TDA's computational constraints, we sample 20 assets per market. We align multi-frequency data (24/7 crypto vs. 5-day equities) using forward-filling (ffill). Following Yen [3] and Kulkarni [4], this standard imputation preserves weekend macro shocks without look-ahead bias, generating discrete time series ready for metric space embedding.

2. Metric Embedding

To capture the evolving structure of the market, we employ a **sliding window** ($W = 60$ days). As this window rolls forward in time, we convert non-stationary prices $P_i(t)$ into log-returns and compute the correlation C_{ij} for all asset pairs *strictly within that timeframe*.

$$C_{ij} = \frac{R_i(t) = \ln(P_i(t)) - \ln(P_i(t-1))}{\sqrt{\sum_{t=1}^W (R_i(t) - \mu_i)^2} \sqrt{\sum_{t=1}^W (R_j(t) - \mu_j)^2}}$$

This correlation is rigorously mapped to a topological distance:

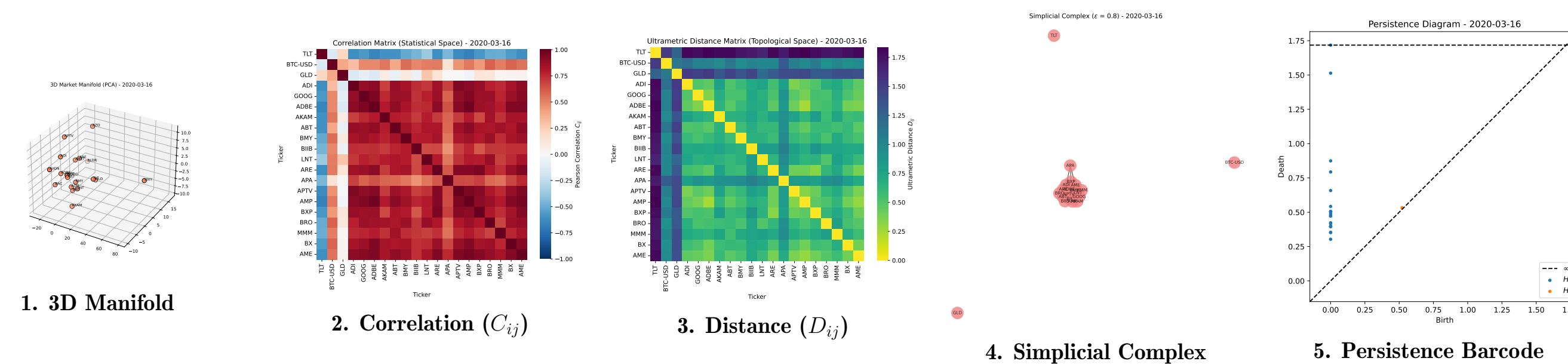
$$D_{ij} = \sqrt{2(1 - C_{ij})}$$

This transformation embeds our assets into a **point cloud**.

Topological Implication: Using the **Vietoris-Rips filtration**, a smaller distance D_{ij} implies highly correlated price movements. When a group of stocks forms a k -simplex at a low filtration threshold, it signifies tightly synchronized, collective market behavior (e.g., sector-wide sell-offs during a crash).

3. Topological Collapse: Case Study (2020-03-16)

By observing the 1-skeleton of the Simplicial Complex during crisis periods (e.g., the $\epsilon = 0.80$ threshold during the COVID Black Monday crash), we observe a distinct geometric transformation across all three key spaces.



Observations:

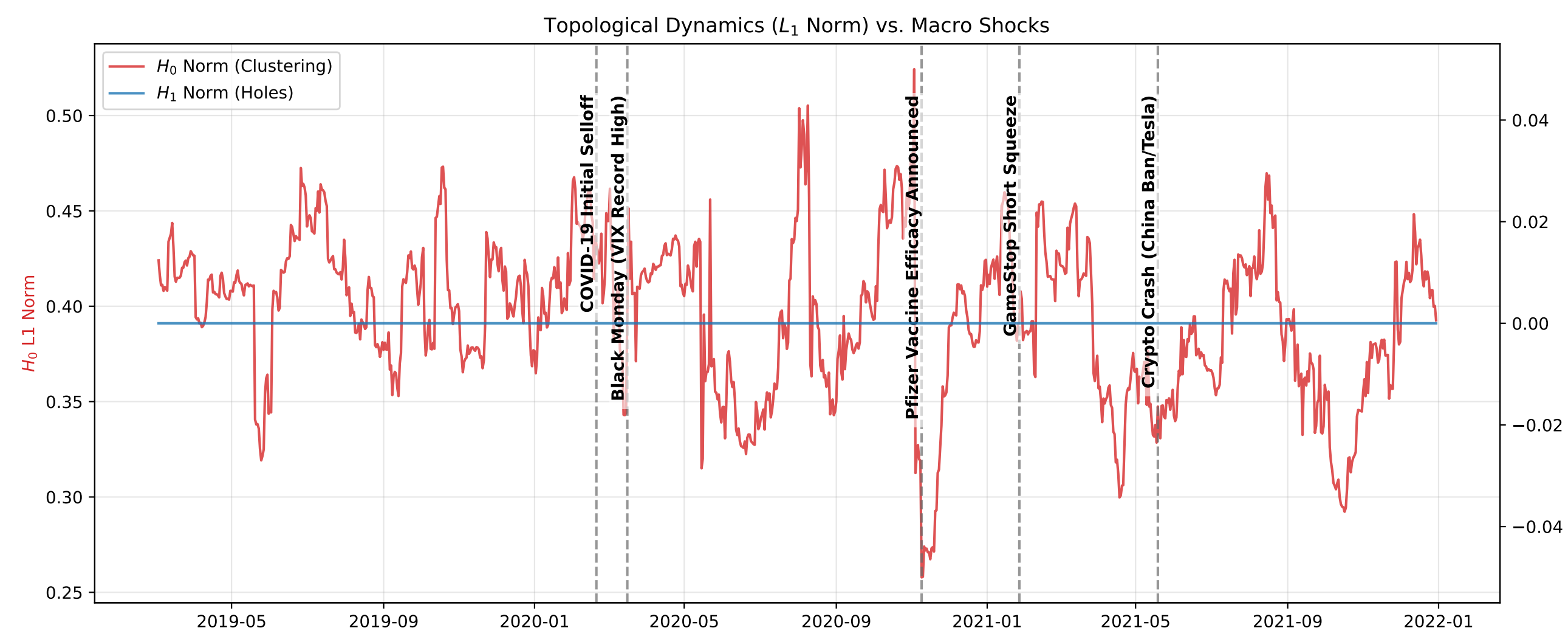
- **High Homogeneity:** The network collapses into a dense clique. Assets move synchronously, filling higher-dimensional simplices and eliminating 1D topological loops (H_1).
- **Outlier Preservation:** Hedging assets (e.g., Gold/NEM, Long Bonds/TLT) remain geometrically distant from the central cluster, validating the topological distance metric.

4. L_1 Norm Time Series vs. Macro Shocks

Drawing upon the methodology presented by Gidea et al [5], we construct Persistence Landscapes and extract their L_1 norm to serve as a single, highly sensitive scalar indicator of market instability.

Mathematical Formulation [5]: To quantify the topological complexity, we transform the persistence barcode intervals $[b_i, d_i]$ into a sequence of Persistence Landscape functions $\lambda_k(t)$, and compute the L_1 norm:

$$||\lambda||_1 = \sum_{k=1}^{k_{max}} \int_0^\infty |\lambda_k(t)| dt$$



Reflecting the COVID-19 "Black Monday": The H_0 curve exhibits a distinct "drop-then-spike" pattern. The sudden drop visualizes absolute panic, where market diversity vanishes as all assets plunge simultaneously. The subsequent sharp rise captures the turbulent aftermath, driven by extreme sector divergence and chaotic market reshuffling.

5. Data Availability & Reproducibility

To ensure full reproducibility of the topological feature extraction and statistical analyses, all data and computational implementations are made publicly accessible.

Open Source Implementation:

- The complete Python codebase, including the **TDAFinancialEngine**, distance matrix computations, and persistence diagram visualizations, is hosted on GitHub.



Scan to access the code repository

github.com/Yun-Qiao11283/TDA

6. Limitations and Future research

Compute constraints currently limit our analysis to a 20-asset subsample, yet our results closely align with foundational full-market studies [Gidea & Katz, 2018]. This suggests that ****Selective market sampling**** can robustly capture systemic risk. Future research will focus on **Optimal Asset Selection**—investigating which market sectors provide the strongest topological signals—and testing the **Stability of Persistence Landscapes** across varying time windows to ensure our early-warning signals are not sensitive to sampling noise.

References

- [1] Otter, N., Porter, M. A., Tillmann, U., Grindrod, P., & Harrington, H. A. (2017). A roadmap for the computation of persistent homology. *EPJ Data Science*, 6(1), 17.
- [2] Rudkin, S., Rudkin, W., & Dłotko, P. (2023). On the topology of cryptocurrency markets. *International Review of Financial Analysis*, 89, 102759.
- [3] Yen, P. T.-W., & Cheong, S. A. (2021). Using topological data analysis (TDA) and persistent homology to analyze the stock markets in Singapore and Taiwan. *Frontiers in Physics*, 9, 572216.
- [4] Kulkarni, S., Pharasi, H. K., Vijayaraghavan, S., Kumar, S., Chakraborti, A., & Samal, A. (2023). Investigation of Indian stock markets using topological data analysis and geometry-inspired network measures. *arXiv preprint arXiv:2311.17016*.
- [5] Gidea, M., & Katz, Y. (2018). Topological data analysis of financial time series: Landscapes of crashes. *Physica A: Statistical Mechanics and its Applications*, 491, 820-834.

Declaration of AI Use

Model: Gemini 3 Flash (Google)

- **Coding:** Assisted in L^AT_EX typesetting, data preprocessing (forward-filling, sliding windows), and implementation via the **giotto-tda** library.